SYSTEMATIC ANALYSIS OF AN UNKNOWN COMPOUND

Functional groups in an unknown organic compound are identified by carrying out the following tests. Confirmatory test is then carried out for each functional group detected. Treat the unknown compound with the following reagents and record all your observations and interferences.

1. Check the solubility in water.
2. Treat with 5% aq. HCL. Observe solubility if it was insoluble in water.
3. Treat with 5% aq. NaOH solution. Observe solubility and the color of the solution. If N is present, see if ammonia is evolved. Heat the solution and observe any changes.
4. Treat with 5% aq. $NaHCO_3$ solution. Observe solubility and test for the evolution of CO_2.
5. Heat with excess soda lime. Test for any gaseous products (NH_3). Collect the
6. Volatile products (phenols, anilines etc.). In a test tube containing alcohol. Carry out the test for phenols and amines on the condensate.
7. Treat with the Brady's reagent .If positive, test for the aldehydes using Fehlings test.
8. Tollen's test or chromic acid test.
9. Add neutral $FeCl_3$ to a solution of the sample.
10. Carry out the ferric hydroxamate test for esters.
11. Test with Br_2/CCl_4 or with Br_2/water.
12. If N is present, carry out the carbylamines test.
13. Lucas test.